

chap 4 section 4.4 :

L'Hospital's Rule: suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a) if:

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ has the form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ then,
 $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ if the limit on the right exist or $\pm\infty$

[1] $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ [2] $0 \cdot \infty$ or $0 \cdot -\infty \rightarrow f \cdot g = \frac{f}{1/g}$ or $\frac{g}{1/f}$

[3] $1^\infty, 0^0, \infty^0 \rightarrow$ taking the natural logarithm

[4] $\infty \cdot \infty \rightarrow$ "توحيد صيغات أو ضرب بالمرافقة"

Ex:-

1- $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \rightarrow \frac{0}{0}$

L.R: $\lim_{x \rightarrow 2} \frac{2x}{1} = \frac{2(2)}{1} = 4$

2- $\lim_{x \rightarrow 0} \frac{-1 - x + e^x}{x^2} \rightarrow \frac{0}{0}$

L.R: $\lim_{x \rightarrow 0} \frac{-1 - x + e^x}{2x} = \frac{0}{0}$

L.R 2: $\lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{e^0}{2} = \frac{1}{2}$

$$3. \lim_{x \rightarrow \frac{\pi}{2}^-} (2x - \pi) \sec x \quad \rightarrow = 0 \cdot \infty$$

$$\hookrightarrow \lim_{x \rightarrow \frac{\pi}{2}^-} (2x - \pi) \frac{1}{\cos x} = \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{2x - \pi}{\cos x} \quad \rightarrow = \frac{0}{0}$$

$$\text{L.R: } \lim_{x \rightarrow \frac{\pi}{2}^-} \frac{\frac{2}{2}}{-\sin x} = \frac{2}{-1} = -2$$

$$4. \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}} \quad \rightarrow = \frac{\infty}{\infty}$$

$$\text{L.R: } \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{3} x^{-2/3}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{3x^{2/3}}}$$

$$= \lim_{x \rightarrow \infty} \frac{3x^{2/3}}{x}$$

$$= 3 \lim_{x \rightarrow \infty} \frac{x^{2/3}}{x} = 3 \lim_{x \rightarrow \infty} \frac{1}{x^{1/3}}$$

$$= 3 \lim_{x \rightarrow \infty} \frac{1}{\sqrt[3]{x}} = \frac{0}{\downarrow}$$

★ any number on ∞ equal zero except $(0, \infty)$.

$$5. \lim_{x \rightarrow 0^+} \left[\frac{1}{e^x - 1} - \frac{1}{x} \right] \quad \rightarrow = \infty - \infty$$

$$\lim_{x \rightarrow 0^+} \frac{x - e^x + 1}{x(e^x - 1)} = \frac{0}{0}$$

$$\text{L.R: } \lim_{x \rightarrow 0^+} \frac{1 - e^x}{e^x - 1 + xe^x} = \frac{0}{0}$$

$$\text{L.R: } \lim_{x \rightarrow 0^+} \frac{-e^x}{e^x + e^x + e^x x}$$

$$= \lim_{x \rightarrow 0^+} \frac{-e^x}{e^x(1 + 1 + x)} = \lim_{x \rightarrow 0^+} \frac{-e^x}{e^x(x + 2)}$$

$$= \lim_{x \rightarrow 0^+} \frac{-1}{(x + 2)} = \frac{-1}{0 + 2} = \frac{-1}{2}$$

* ملاحظات مهم : في حالات $0 \cdot \infty$:

(ا) إذا كان عندك poly مع الـ $\ln(x)$ - نضع $\ln(x)$ في البسط و مقرب
الحدودية في المقام .

(ب) إذا كان (inverse . trig) مع $\ln(x)$ - نضع (inverse . trig) في
البسط وفي المقام نضع مقرب $\ln(x)$.

$$6 - \lim_{x \rightarrow 0^+} x \ln x$$

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}}$$

$$L.R : \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\frac{-1}{x^2}} = - \lim_{x \rightarrow 0^+} \frac{x^2}{x} = - \lim_{x \rightarrow 0^+} x = 0$$

$$7 - \lim_{x \rightarrow 0^+} x^x$$

$$(y = x^x)$$

$$\ln y = \ln x^x \rightarrow \ln y = x \ln x$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-1/x^2} = - \lim_{x \rightarrow 0^+} x = 0$$

$$\lim_{x \rightarrow 0^+} \ln y = 0 \rightarrow \lim_{x \rightarrow 0^+} e^{\ln y} = e^0$$

$$\lim_{x \rightarrow 0^+} y = 1 \rightarrow \lim_{x \rightarrow 0^+} x^x = 1$$

Section (7.1) :-

integration by parts: $\int u dv = uv - \int v du$

(or) $\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$

Cases :-

[1] $\int x^n \left(\begin{array}{c|c} \sin x & a^x \\ \cos x & e^x \\ \tan x & \end{array} \right) dx \rightarrow u = x^n$
 $dv = ()$

[2] $\int x^n \left(\begin{array}{c|c} \sin^{-1} x & \ln x \\ \cos^{-1} x & \log x \\ \tan^{-1} x & \end{array} \right) dx \rightarrow u = () , dv = x^n$

[3] $\int \left(\begin{array}{c|c} \sin x & a^x \\ \cos x & e^x \\ \tan x & \end{array} \right) \cdot \left(\begin{array}{c|c} \sin^{-1} x & \ln x \\ \cos^{-1} x & \log x \\ \tan^{-1} x & \end{array} \right) dx \rightarrow \text{repeated integral}$

(+ هيا لا يفرق u و dv ، يكون هناك تكرار لذا يجب الانتباه إلى
 نفس العلاقة .)

ii) * proof of (I.B.P) :-

$$(uv)' = uv' + u'v$$

$$= u dv + du v$$

$$uv = \int u dv + \int v du$$

$$\int u dv = uv - \int v du$$

Q. 10

Let f and g be two differentiable functions. prove that

$$\int f(x) g'(x) dx = f(x) g(x) - \int g(x) f'(x) dx.$$

→ ∴ f and g are differentiable.

the product rule gives:-

$$\frac{d}{dx} (f(x) g(x)) = f'(x) g(x) + g'(x) f(x)$$

integrating both sides:

$$\begin{aligned} f(x) g(x) &= \int f'(x) g(x) dx + \int g'(x) f(x) dx \\ \int f'(x) g(x) dx &= f(x) g(x) - \int g'(x) f(x) dx \end{aligned}$$

Ex:

1. $\int \ln x dx$

$$u = \ln x \quad dv = dx$$

$$du = \frac{1}{x} dx \quad v = x$$

$$\begin{aligned} I &= x \ln x - \int x \cdot \frac{1}{x} dx \\ &= x \ln x - \int dx \end{aligned}$$

$$= x \ln x - x + C$$

7- let f and f' are cont. on $[0, 1]$, $f(1) = -1$ and $\int_0^1 f(x) dx = 2$. Find $\int_0^1 x f'(x) dx$.

$$I = \int_0^1 x f'(x) dx$$

$$u = x \quad dv = f'(x) dx$$

$$du = dx \quad v = f(x)$$

$$\begin{aligned} I &= [x f(x)]_0^1 - \int_0^1 f(x) dx \\ &= [(1) f(1) - (0) f(0)] - \int_0^1 f(x) dx \\ &= (-1) - (2) \\ &= -3 \end{aligned}$$

* عندما يكون التكامل $\int e^{(\dots)} dx$ $\leftarrow$ نستخدم $t = (\dots)$ ونجرب ان $(\dots)$ هي ايع في dx او $(x) < f'(x)$ في $t = (\dots)$ ومن ثم نجرب I.B.P.

Ex:

$$\int e^{\sqrt{x}} dx$$

$$t = \sqrt{x} \quad \text{نضع}$$

$$t^2 = x \quad \text{نربع}$$

$$2t dt = dx \quad \text{نسوّي}$$

$$I = \int e^t 2t dt \quad \text{I.B.P}$$

$$u = t \quad dv = e^t dt$$

$$du = dt \quad v = e^t$$

$$I = t e^t - \int e^t dt$$

$$= t e^t - e^t + C = 2[\sqrt{x} e^{\sqrt{x}} - e^{\sqrt{x}}] + C$$

Section (7.2):-

Trigonometric integrals

$$\int \sin^n x \cos^m x \, dx \quad \text{or} \quad \int \sinh^n x \cosh^m x \, dx$$

□ if (n) or (m) is odd:

$$\sin^2 x = 1 - \cos^2 x$$

$$\cos^2 x = 1 - \sin^2 x$$

□ if both (n) and (m) are even:

$$(x2) \quad \sin^2 x = \frac{1}{2} (1 - \cos 2x) \quad (x2)$$

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

$$2\sin^2 x = 1 - \cos(2x)$$

$$2\cos^2 x = 1 + \cos(2x)$$

$$\left[\begin{array}{l} 1 - \cos(\theta) = 2\sin^2\left(\frac{\theta}{2}\right) \\ 1 + \cos(\theta) = 2\cos^2\left(\frac{\theta}{2}\right) \end{array} \right]$$

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x)$$

$$\star \quad \operatorname{csch}^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csch}^2 x$$

$$\sin(x) \cos(y) = \frac{1}{2} [\sin(x+y) + \sin(x-y)]$$

$$\cos(x) \cos(y) = \frac{1}{2} [\cos(x+y) + \cos(x-y)]$$

$$\sin(x) \sin(y) = \frac{1}{2} [\cos(x-y) - \cos(x+y)]$$

$$\cos(-x) = \cos x, \quad \sin(-x) = -\sin x, \quad \tan(-x) = -\tan x$$

$$\int \tan^n x \sec^m x dx \text{ or } \int \cot^n x \csc^m x dx$$

1] if (n) is odd (tan or cot):-

$$\tan^2 x = \sec^2 x - 1, \cot^2 x = \csc^2 x - 1$$

2] if (m) is even (sec or csc):-

$$\sec^2 x = \tan^2 x + 1, \csc^2 x = \cot^2 x + 1$$

3] if (n) is even and (m) is odd:-

I.B.P

$$\int \tan^3 x dx$$

$$I = \int \tan^2 x \tan x dx$$

$$= \int (\sec^2 x - 1) \tan x dx$$

$$u = \sec x \quad du = \sec x \tan x dx$$

$$= \int \sec^2 x \tan x - \int \tan x dx$$

$$u = \tan x, du = \sec^2 x dx$$

$$= \int u du - \int \tan x dx$$

$$= \frac{u^2}{2} - \ln |\sec x| + c$$

$$= \frac{\tan^2 x}{2} - \ln |\sec x| + c$$

$$\int \tan^3 x \sec^{5/2} x dx$$

$$= \int \tan^2 x \tan x \sec x \sec^{3/2} x dx$$

$$= \int (\sec^2 x - 1) \tan x \sec x \sec^{3/2} x dx$$

$$u = \sec x \quad du = \sec x \tan x dx$$

$$= \int (u^2 - 1) u^{3/2} du = \int u^{7/2} - u^{3/2} du$$

$$= \frac{2}{9} u^{9/2} - \frac{2}{5} u^{5/2} + c, u = \sec x$$

section (7.3):-

Trigonometric substitution

$$1) \sqrt{a^2 - x^2} \rightarrow x = a \sin \theta$$

$$2) \sqrt{a^2 + x^2} \rightarrow x = a \tan \theta$$

$$3) \sqrt{x^2 - a^2} \rightarrow x = a \sec \theta$$

يمكن استخدام جميع
النظريات الثلاث
فقط

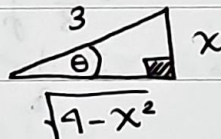
$$- \int \frac{\sqrt{9 - x^2}}{x^2} dx$$

$$\sqrt{9 - x^2} = \sqrt{(3)^2 - x^2}, \quad a = 3$$

$$x = 3 \sin \theta \rightarrow \sin \theta = \frac{x}{3} \rightarrow \theta = \sin^{-1} \left(\frac{x}{3} \right)$$

$$dx = 3 \cos \theta d\theta$$

$$\sqrt{9 - x^2} = 3 \cos \theta$$



$$I = \int \frac{3 \cos \theta \cdot 3 \cos \theta d\theta}{9 \sin^2 \theta} = \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta$$

$$= \int \cot^2 \theta d\theta = \int (\csc^2 \theta - 1) d\theta$$

$$= \int \csc^2 \theta d\theta - \int 1 d\theta$$

$$= -\cot \theta - \theta + C$$

$$= -\frac{\sqrt{9 - x^2}}{x} - \sin^{-1} \left(\frac{x}{3} \right) + C$$

completing the square Formula:

$$ax^2 + bx = (cx + d)^2 + e \quad (a > 0)$$

$$c = \sqrt{a}, \quad d = \frac{b}{2\sqrt{a}}, \quad e = -(d)^2$$

* إذا لم يكن معامل (x^2) موجباً، وإذا كان سالباً، فاضرب السالب عامل مشتركاً.

• غالباً نفكر في استخدام المربع الكامل إذا كان هناك جدورية من الدرجة الثانية (أو الرابعة) في السالب أو المقام تحت الجذر التربيعي.

$$-\int \frac{dx}{\sqrt{x^2 + 8x + 25}}$$

or $(8 \times \frac{1}{2} = 4 \rightarrow (4)^2 \rightarrow 16 \rightarrow$ الجمعها
والمرحما

$$a = 1 \quad b = 8 \quad c = \sqrt{1} = 1 \quad d = \frac{8}{2} = 4$$

$$e = -(4)^2 = -16$$

$$x^2 + 8x = (x + 4)^2 + (-16) = (x + 4)^2 - 16$$

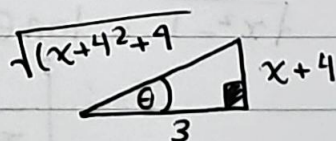
$$\begin{aligned} \sqrt{x^2 + 8x + 25} &= \sqrt{(x + 4)^2 - 16 + 25} = \sqrt{(x + 4)^2 + 9} \\ &= \sqrt{(x + 4)^2 + (3)^2} \end{aligned}$$

$$a = 3, \quad x + 4 = 3 \tan \theta$$

$$\tan \theta = \frac{x + 4}{3} \rightarrow \theta = \tan^{-1} \left(\frac{x + 4}{3} \right)$$

$$x = 3 \tan \theta - 4 \rightarrow dx = 3 \sec^2 \theta d\theta$$

$$\sqrt{(x + 4)^2 + 9} = 3 \sec \theta$$



$$I = \int \frac{3 \sec^2 \theta d\theta}{3 \sec \theta} = \int \sec \theta d\theta$$

$$= \ln | \sec \theta + \tan \theta | + C$$

$$= \ln \left| \frac{\sqrt{(x + 4)^2 + 9}}{3} + \frac{x + 4}{3} \right| + C$$

Section (7.4):-

assume we have $\frac{P}{Q} = \frac{\text{poly}}{\text{poly}}$

1) if $\deg(f) \geq \deg(g) \rightarrow$ 1) long division.
2) factorize.

2) if $\deg(f) < \deg(g) \rightarrow$ 1) factorize.

ex:

$$\frac{1}{x^3 (x+1)^2} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x+1} + \frac{E}{(x+1)^2}$$

$$\frac{x}{(x^2-1)(x^2+4)^2} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+4} + \frac{Ex+F}{(x^2+4)^2}$$

$\downarrow$ $\downarrow$ $\swarrow$ $\searrow$
 $(x-1)$ $(x+1)$ مقسوم عليه مقسوم عليه

$$\frac{x-5}{(x+3)(x^2+1)(x^2-4)} = \frac{A}{x+3} + \frac{Bx+C}{x^2+1} + \frac{D}{x-2} + \frac{E}{x+2}$$

$\swarrow$ $\searrow$
 مقسوم عليه مقسوم عليه

substitution: $\frac{\text{poly}}{\text{poly}} \rightarrow \frac{f}{g} \rightarrow \text{not poly}$

Forexample: -

$$\int \frac{3e^{4x} + 3e^{3x} - 5e^{2x} + e^x - 1}{e^{2x} + e^x - 2} e^x dx$$

not poly
not poly

$$u = e^x \quad du = e^x dx$$

$$\int \frac{3u^4 + 3u^3 + 5u^2 + u - 1}{u^2 + u - 2} du \rightarrow \frac{\text{poly}}{\text{poly}}$$

* في حالة وجود جذور مختلفة ويكون ما تحت الجذر متماثل نظريا:

$$u = x^{1/n} = \sqrt[n]{x} \quad \text{where } n : \text{L.C.M. (العامل المشترك الأصغر)}$$

$$4 - \int \frac{dx}{x(\sqrt{x} + \sqrt[4]{x})}$$

$$u = \sqrt[4]{x}$$

$$u^4 = x \rightarrow \sqrt{x} = \sqrt[4]{u^4} = u^2$$

$$dx = 4u^3 du$$

$$\int \frac{4u^3 du}{u^4(u^2 + u)} = \int \frac{4}{u(u^2 + u)} du = \int \frac{4}{u^2(u + 1)} du$$

$$\frac{4}{u^2(u+1)} = \frac{A}{u} + \frac{B}{u^2} + \frac{C}{u+1}$$

$$1 = A(u)(u+1) + B(u+1) + C(u^2)$$

$$\boxed{u=0} \rightarrow \boxed{1=B}$$

$$1 = Au^3 + Au + \frac{1}{u^2} + \frac{1}{u} + Cu^2 + Du^2$$

$$1 = u^3(A+C) + u^2(1+D) + Au + 1$$

$$A+C=0 \rightarrow A=-C$$

$$1+D=0 \rightarrow \boxed{D=-1}$$

$$\boxed{A=0} \rightarrow \boxed{C=0}$$

$$\int \frac{4}{u^2(u+1)} du = 4 \left[\int \frac{1}{u^2} du - \int \frac{1}{u+1} du \right]$$

$$= 4 \left[\frac{u^{-1}}{-1} - \tan^{-1}(u) + C \right], \quad u = \sqrt[4]{x}$$

$$= \frac{-4}{\sqrt[4]{x}} - 4 \tan^{-1}(\sqrt[4]{x}) + C$$

Section (7.8) :-

Improper Integrals : 1) Integrals over ∞ interval.

2) f has ∞ discontinuity on $[a, b]$.

□ • if $\int_a^t f(x) dx$ exists for every number $t \geq a$, then
$$\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx.$$

• if $\int_t^b f(x) dx$ exists for every number $t < b$, then
$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx.$$

→ the improper integrals $\int_a^{\infty} f(x) dx$ and $\int_{-\infty}^b f(x) dx$ are called convergent if the corresponding limit exists, and divergent if the limit does not exist.

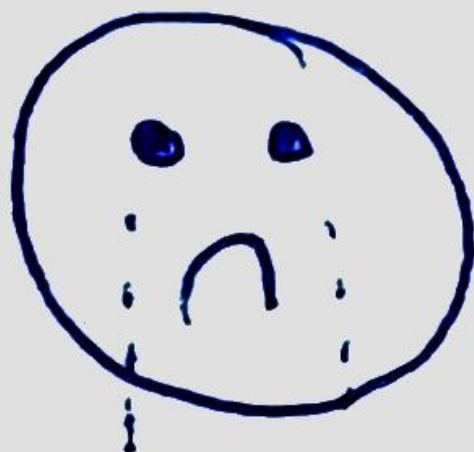
• If both $\int_a^{\infty} f(x) dx$ and $\int_{-\infty}^a f(x) dx$ are convergent, then we define :
$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx.$$

$$- \int_1^{\infty} \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx$$

$$= \lim_{t \rightarrow \infty} [\ln|x|]_1^t = \lim_{t \rightarrow \infty} [\ln|t| - \cancel{\ln|1|}]$$

$$= \ln(\infty) = \infty \quad \underline{\text{Divergent}}$$

second
mid



Section (11.1) :-

An infinite sequences:

$\hookrightarrow$ is a function whose domain is the set of positive integers: $a_1, a_2, a_3, \dots, a_{n-1}, a_n, a_{n+1} = \{a_n\}_{n=1}^{\infty}$.

examples:

1- $2, 4, 6, 8, \dots, 2n, \dots = \{2n\}_{n=1}^{\infty}$

$\rightarrow$ sequence of even number, increasing sequence.

2- $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots, \frac{1}{n}, \dots$

$\rightarrow$ decreasing sequence, goes to zero.

3- $1, 3, 5, 7, 9, \dots, 2n-1, \dots = \{2n-1\}_{n=1}^{\infty}$

$\rightarrow$ sequence of odd numbers, increasing sequence.

4- $\left\{\frac{n-1}{n}\right\}_{n=1}^{\infty}, 0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \frac{n-1}{n}, \dots$

$\rightarrow$ increasing sequence, goes to one.

5- $\{5\}_{n=1}^{\infty} = 5, 5, 5, 5, \dots, 5, \dots$

$\rightarrow$ constant sequence, goes to 5.

6- $\left\{(-1)^{n+1} \frac{1}{n}\right\}_{n=1}^{\infty} = 1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}, \dots, (-1)^{n+1} \frac{1}{n}, \dots$

$\rightarrow$ neither decreasing nor increasing sequence.

7- the sequence with $a_1 = 1, a_{n+1} = \frac{a_n}{n+1}$

$$a_1 = 1, \quad a_2 = \frac{a_1}{1+1} = \frac{1}{2}$$

$$a_3 = \frac{a_2}{2+1} = \frac{\frac{1}{2}}{3} = \frac{1}{6}$$

$$a_4 = \frac{a_3}{3+1} = \frac{\frac{1}{6}}{4} = \frac{1}{24}$$

$\therefore$ Decreasing sequence, goes to zero.

- n Factorial ($n!$):

$$n! = n \times (n-1) \times (n-2) \times (n-3) \dots 3 \times 2 \times 1.$$

$$\rightarrow 0! = 1, \quad \infty! = \infty$$

- convergence of sequences:-

1) If $\lim_{n \rightarrow \infty} a_n = L$ (exist) we say that $\{a_n\}$ Converge.

2) If $\lim_{n \rightarrow \infty} a_n = \text{fails to exist } (\infty)$ we say that $\{a_n\}$ diverge.

example:-

1- $\lim_{n \rightarrow \infty} 2n = 2(\infty) = \infty \rightarrow \{2n\}$ is div.

2- $\lim_{n \rightarrow \infty} \frac{1}{n} = \frac{1}{\infty} = 0 \rightarrow \left\{\frac{1}{n}\right\}$ is conv.

3- $\lim_{n \rightarrow \infty} (2n-1) = 2(\infty)-1 = \infty \rightarrow a_n$ is div.

4- $\lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right) = \frac{n}{n} - \frac{1}{n} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1 - \frac{1}{\infty}$
 $= 1 - 0 = 1 \rightarrow a_n$ is conv.

5- $\lim_{n \rightarrow \infty} 5 = 5 \rightarrow a_n$ is conv.

Theorem: If $\lim_{n \rightarrow \infty} |a_n| = 0$ then $\lim_{n \rightarrow \infty} a_n = 0$.

6- $\left\{(-1)^{n+1} \frac{1}{n}\right\}_{n=1}^{\infty}$

هذا متسلسلة متناوبة
لأنها تتناوب بين الموجب والسالب
والحد هو 0

$$a_n = (-1)^{n+1} \frac{1}{n} \rightarrow |a_n| = \left|(-1)^{n+1} \frac{1}{n}\right| = \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} \frac{1}{n} = \frac{1}{\infty} = 0$$

$$\therefore \lim_{n \rightarrow \infty} a_n = 0 \rightarrow \{a_n\} \text{ is conv.}$$

$$7 - \lim_{n \rightarrow \infty} \frac{1}{n!} = \frac{1}{\infty!} = \frac{1}{\infty} = 0 \rightarrow a_n \text{ is conv.}$$

$$8 - \{(-1)^{n+1}\}$$

$$\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} |(-1)^{n+1}| = 1 \neq 0$$

$$a_n = (-1)^{n+1} = \begin{cases} 1, & \text{if } n \text{ is odd.} \\ -1, & \text{if } n \text{ is even.} \end{cases}$$

$$\lim_{n \rightarrow \infty} a_n = \text{DNE} \quad \because a_n \text{ is div.}$$

↳ because it has two different limits.

- Theorem (sandwich theorem):

Let $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ be a sequences of real numbers. If $\{a_n\} \leq \{b_n\} \leq \{c_n\}$ for all n beyond some index N , and if

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L \quad \text{then, } \lim_{n \rightarrow \infty} b_n = L.$$

- Theorem:

Let $\{a_n\}$ and $\{b_n\}$ be sequences of real number and let A and B be real numbers. If $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$. then,

- $\lim_{n \rightarrow \infty} (a_n \pm b_n) = A \pm B$.
- $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = AB$.
- $\lim_{n \rightarrow \infty} (k \cdot b_n) = kB$. (k is any real number).
- $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}$, $B \neq 0$.

Basic limits :-

$$① \lim_{n \rightarrow \infty} \frac{\ln(n)}{n} = 0$$

$$② \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$$

$$③ \lim_{n \rightarrow \infty} (x)^{\frac{1}{n}} = 1, \quad x > 0$$

$$④ \lim_{n \rightarrow \infty} (x)^n = \begin{cases} 0, & -1 < x < 1 \\ 1, & x = 1 \\ \infty, & x > 1 \\ \text{DNE}, & x \leq -1 \end{cases}$$

$$⑤ \lim_{n \rightarrow \infty} \frac{(x)^n}{n!} = 0, \text{ for any } x.$$

→ (In general, $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x, \quad x \in \mathbb{R}$)

$$⑥ \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x, \text{ for any } x.$$

$$⑦ \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{k}{n}\right)}{\left(\frac{k}{n}\right)} = 1, \text{ where } k \in \mathbb{R}, \quad \frac{k}{n} \rightarrow 0$$

$$⑧ \lim_{n \rightarrow \infty} \frac{\tan\left(\frac{k}{n}\right)}{\left(\frac{k}{n}\right)} = 1$$

$$⑨ \lim_{n \rightarrow \infty} \frac{1 - \cos\left(\frac{k}{n}\right)}{\left(\frac{k}{n}\right)} = 0$$

section (11.2) :-

Infinite series :-

↳ the sum of infinite seq.

* Conv. of series :

$$\text{let } S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$S_n = a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n$$

$$\rightarrow S_{n-1} = a_1 + a_2 + a_3 + \dots + a_{n-1} \quad \uparrow$$

$$S_n = S_{n-1} + a_n$$

$$* a_n = S_n - S_{n-1} *$$

$$- \lim_{n \rightarrow \infty} S_n = L \rightarrow \{S_n\} \text{ Conv. } \rightarrow \sum_{n=1}^{\infty} a_n = L \text{ (Conv.)}$$

$$- \lim_{n \rightarrow \infty} S_n = \left(\begin{matrix} \text{DNE} \\ \pm \infty \end{matrix} \right) \rightarrow \{S_n\} \text{ div. } \rightarrow \sum_{n=1}^{\infty} a_n \text{ (div.) no sum.}$$

→ لا تقبل جمعاً لا نهائي

① Telescoping series (T.S) :-

$$1 - () - ()$$

$$2 - \frac{()}{()} \rightarrow \text{P.F. det} \rightarrow () - ()$$

Geometric series (G.S):-

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots$$

- a is the first term, $a \neq 0$.

- r is the ratio.

- The G.S $\sum_{n=1}^{\infty} ar^{n-1}$ is conv. If $|r| < 1$, $\text{Sum} = \frac{a}{1-r}$.

- The G.S $\sum_{n=1}^{\infty} ar^{n-1}$ is div. If $|r| \geq 1$, no sum.

أولاً في الـ G.S عندنا يكون لدينا a أو a هو أول حد أو a هو الحد الأول.
أو a هو الحد الأول أو a هو الحد الأول.

$$\textcircled{1} \sum_{n=0}^{\infty} (-1)^n \frac{5}{4^n} = \sum_{n=0}^{\infty} \frac{5(-1)^n}{4^n}$$

$$= \sum_{n=0}^{\infty} 5 \left(\frac{-1}{4} \right)^n$$

$$a = 5 \quad r = -\frac{1}{4}$$

$$|r| = \left| -\frac{1}{4} \right| = \frac{1}{4} < 1 \quad (\text{Conv.})$$

$$S = \frac{a}{1-r} = \frac{5}{1+\frac{1}{4}} = \frac{5}{\frac{5}{4}} = \frac{4 \times 5}{5} = 4.$$

Show that sum of G.S if it is conv. $S = \frac{a}{1-r}$

$$a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots = \sum_{n=1}^{\infty} ar^{n-1}$$

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-1}$$

$$rS_n = ar + ar^2 + ar^3 + \dots + ar^{n-1} + ar^n$$

$$S_n - rS_n = a - ar^n$$

$$S_n(1-r) = a(1-r^n) \rightarrow S_n = \frac{a(1-r^n)}{(1-r)}$$

Section (11.8):-

Integral test:

let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms, and let $f(x)$ be a function obtained by replacing n by x in the formula of a_n , If:

① $f(x)$ is positive.

② $f(x)$ is cont. on $[N, \infty)$.

③ $f(x)$ is decreasing for all $n \geq N$.

Then the series $\sum_{n=1}^{\infty} a_n$ and $\int_N^{\infty} f(x) dx$ are both either Conv. or div.

If ♥ is Conv. $\rightarrow$ ☼ Conv.

If ♥ is div. $\rightarrow$ ☼ div.

* أفتي غالباً في هذا الاختبار - لأنني نكامل الدالة بكل سهولة.

① لو فرضنا $u = (x)$ ووجد المشتقة أو

جزء منها بالـ x .

Section (11.4):-

Comparison test and limit Comparison test.

- The Comparison test:-

let $\sum a_n$, $\sum C_n$, and $\sum d_n$ be series with positive terms, suppose that for some integer N .

$$d_n \leq a_n \leq C_n$$

$\downarrow$ $\swarrow$ $\searrow$ $\downarrow$
 Div. Conv. Conv.

→ If $\sum C_n$ is Converges, then $\sum a_n$ also Converges.

→ If $\sum d_n$ is diverges, then $\sum a_n$ is also diverges.

في ال (C.T) :-

Conv. 1- إذا انتهت بمضاد ① تكس الب + ② تصغر المقام

Div. 2- إذا انتهت بإرأ: ① تصغر الب + ② تكس المقام

$$2^n > n$$

$$\frac{1}{n!} < \frac{1}{2^{n-1}}$$

غالبًا إذا رأينا: $\sin(n)$ أو $\cos(n)$ نفكر باستخدام ال (C.T.)

أو إذا رأينا: $n!$ نستخدم ال (C.T.) أو (R.T.)

Limit Comparison test (L.C.T):-

Suppose that $a_n > 0$ and $b_n > 0, \forall n \geq N$

① If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ are both Conv. or both div.

② If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ Conv., then $\sum a_n$ Conv.

③ If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ div., then $\sum a_n$ div.

← غالباً کیفیت اختیار از b_n : آبرو آقون در بی البیه
آبرو آقون در بی البیه

غالباً إذا رأينا: $\sin(\frac{1}{n}), \cos(\frac{1}{n})$ نستعمل هذا الاختبار
نستعمل مع $(n!)$ أو $\sin(n)$ أو $\cos(n)$.

$$\textcircled{1} \sum_{n=1}^{\infty} \left(\frac{1}{n \sqrt[3]{n^3}} \right) = a_n$$

$$b_n = \frac{1}{n} \text{ Div. H.D.}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1}{n \sqrt[3]{n^3}} \cdot n$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\sqrt[3]{n^3}} = 1 > 0$$

$$\left| \lim_{n \rightarrow \infty} \sqrt[3]{n^3} = \lim_{n \rightarrow \infty} (\sqrt[3]{n})^3 = (1)^3 = 1 \text{ Basic limits.} \right.$$

$\therefore \sum b_n$ Div By H.D.

$\therefore \sum a_n$ Div by L.C.T.

Section (11.6):-

(ratio and root test):

- The ratio test (R.T):-

Let $\sum a_n$ be a series with +ve terms ($a_n \geq 0, \forall n \geq N$).

And suppose that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \rho$, Then:-

1- The series $\sum a_n$ Conv. if $\rho < 1$.

2- The series $\sum a_n$ div. if $\rho > 1$.

→ The test is inconclusive (fail) if $\rho = 1$.

- The n-th root test:-

Let $\sum a_n$ be a series with +ve terms ($a_n \geq 0, \forall n \geq N$).

And suppose that $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \rho$

1- The series $\sum a_n$ Conv. if $\rho < 1$.

2- The series $\sum a_n$ div. if $\rho > 1$.

→ The test is inconclusive (fail) if $\rho = 1$.

$$\textcircled{2} \sum_{n=1}^{\infty} (-1)^n \frac{2n+1}{3n+2}$$

$$a_n = \frac{2n+1}{3n+2}$$

$$\square \lim_{n \rightarrow \infty} \frac{2n+1}{3n+2} = \lim_{n \rightarrow \infty} \frac{\frac{2n}{n} + \frac{1}{n}}{\frac{3n}{n} + \frac{2}{n}} = \frac{2}{3} \neq 0$$

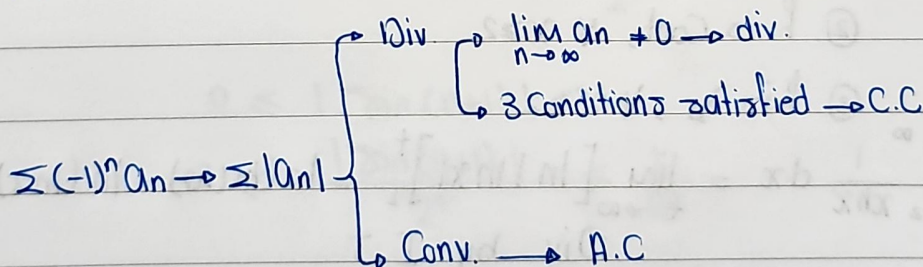
$\therefore \sum (-1)^n \frac{2n+1}{3n+2}$ is div. by n-th term test of div.

→ Absolutely Convergent (A.C.):

If the series of the absolute value $\sum |a_n|$ conv. then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is called absolutely convergent.

→ Conditionally Convergent (C.C.):

If the alternating series is not absolutely convergent but satisfies the three conditions of the theorem then the series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is called conditionally convergent.



Section (11.8) :-

(Alternating series test)

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n+1} a_n + \dots$$

$$(\text{or}) \sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + a_4 + \dots + (-1)^n a_n + \dots$$

The alternating series test (A.S.T) :-

the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ conv. if the following three conditions are satisfied :-

① $\lim_{n \rightarrow \infty} a_n = 0$

② the a_n 's are all positive.

③ the positive a_n 's are decreasing, that is $a_n \geq a_{n+1}$,
 $\forall n \geq N$ for some integer N .

① $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$

$$a_n = \frac{1}{n}$$

① $\lim_{n \rightarrow \infty} \frac{1}{n} = \frac{1}{\infty} = 0$

② $a_n > 0, \forall n \geq 1$

③ $f(x) = \frac{1}{x}$, $f'(x) = -\frac{1}{x^2}$, $\therefore f$ is $\downarrow$

so a_n is $\downarrow$ for all $n \geq 1$

$\therefore \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$ is Conv. by A.S.T

Estimating sums:

(Alternating series estimation):

If $s = \sum (-1)^{n+1} b_n$ is the sum of alternating series that satisfies:

① b_n 's are all +ve.

② $b_n \geq b_{n+1}$, $\forall n \geq N$.

③ $\lim_{n \rightarrow \infty} b_n = 0$

Then $|R_n| = |s - s_n| \leq b_{n+1}$

→ this theorem for Conv. alternating series only!!

① $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$, ($|\text{error}| < 0.01$)

$\sum \frac{(-1)^{n+1}}{n^2}$ is Conv. by A.S.T

$b_n = \frac{1}{n^2}$

$b_{n+1} < \frac{1}{100}$

$\frac{1}{(n+1)^2} < \frac{1}{100}$

$(n+1)^2 > 100$, $\forall n \geq 10$

$\therefore \forall n \geq 10$, $|R_n| = |s - s_n| \leq 0.01$

Section (11.8):-

The power series :

$$\sum_{n=0}^{\infty} a_n (x-c)^n \quad \text{or} \quad \sum_{n=0}^{\infty} a_n x^n, \quad x \neq 0$$

center

(x) ← تعاملو كتابت

$$\frac{b-a}{2} = \text{radius of conv} \quad \leftarrow \text{هدفنا إيجاد}$$

$$\text{interval of conv} \quad \leftarrow$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad \text{ratio test} \quad \leftarrow \text{كيفية استخدام}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \quad n\text{-th root test} \quad \leftarrow$$

□ سوف يظهر لنا $-R < x < R$

□ لازم ندرس إذا x عند الحواف

(a,b) $\leftarrow$ إذا كانت Div $\leftarrow$ قوس مفتوح

$\leftarrow$ إذا كانت Conv $\leftarrow$ قوس مغلق

power series summary

① If $x \in I = [a,b], (a,b), [a,b), (a,b]$

$$R = \frac{b-a}{2}, \quad \lim_{n \rightarrow \infty} = 1$$

② If $x = I = a$

$$R = 0, \quad \lim_{n \rightarrow \infty} = \infty$$

③ If $x \in (-\infty, \infty)$

$$R = \infty, \quad \lim_{n \rightarrow \infty} = 0$$

Section (11.9):-

(Representation of functions as power series
in power of x)

1- $\frac{1}{1+u}$, $\sum_{n=0}^{\infty} (-1)^n u^n$, $|u| < 1$, $I = (-1, 1)$, $R = 1$

2- $\frac{1}{1-u}$, $\sum_{n=0}^{\infty} u^n$, $|u| < 1$, $I = (-1, 1)$, $R = 1$

3- $\ln(1+u)$, $\sum_{n=0}^{\infty} (-1)^n \frac{u^{n+1}}{n+1}$, $I = (-1, 1]$, $R = 1$

4- $\tan^{-1} u$, $\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n+1}}{2n+1}$, $I = [-1, 1]$, $R = 1$

5- e^u , $\sum_{n=0}^{\infty} \frac{u^n}{n!}$, $I = \mathbb{R}$, $R = \infty$

6- $\sinh u$, $\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n+1}}{(2n+1)!}$, $I = \mathbb{R}$, $R = \infty$

7- $\cosh u$, $\sum_{n=0}^{\infty} \frac{u^{2n}}{(2n)!}$, $I = \mathbb{R}$, $R = \infty$

8- $\cos u$, $\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n}}{(2n)!}$, $I = \mathbb{R}$, $R = \infty$

9- $\cosh u$, $\sum_{n=0}^{\infty} \frac{u^{2n}}{(2n)!}$, $I = \mathbb{R}$, $R = \infty$

recall :-

$$1- \sin x \cos x = \frac{1}{2} \sin(2x)$$

$$2 \sin x \cos x = \sin(2x)$$

$$2- \cos^2 x = \frac{1}{2} (1 + \cos 2x)$$

$$3- \sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$4- \cosh^2 x = \frac{1}{2} (\cosh(2x) + 1)$$

$$5- \sinh^2 x = \frac{1}{2} (\cosh(2x) - 1)$$

$$6- \sinh x \cosh x = \frac{1}{2} \sinh(2x)$$

$$7- a^u = e^{u \ln a}$$

$$\textcircled{1} f(x) = \frac{x}{4+x^2}$$

$$\frac{1}{1+u} = \sum_{n=0}^{\infty} (-1)^n u^n$$

$$\frac{1}{4+x^2} = \frac{1}{4(1+\frac{x^2}{4})} = \frac{1}{4} \cdot \frac{1}{1+(\frac{x^2}{4})}$$

$$= \frac{1}{4} \sum_{n=0}^{\infty} (-1)^n \left(\frac{x^2}{4}\right)^n$$

$$\left|\frac{x^2}{4}\right| < 1$$

$$|x^2| < 4$$

$$|x| < 2$$

$$I = (-2, 2)$$

$$R = \frac{2+2}{2} = 2$$

$$\begin{aligned} \frac{1}{4+x^2} &= \frac{1}{4} \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{4^n} \\ x(x) \quad \frac{x}{4+x^2} &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{4^{n+1}} \end{aligned}$$

Section (11.10):-

Taylor and Maclaurin series:-

→ The Taylor series generated by the function f at the point a is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x-a)^n}{n!}$$
$$= f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f^{(n)}(a)(x-a)^n}{n!} + \dots$$

→ If $(a=0)$ we get the Maclaurin series that is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)(x)^n}{n!}$$
$$= f(0) + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!} + \dots + \frac{f^{(n)}(0)x^n}{n!} + \dots$$

Remark 8-

1. $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$

2. $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$

3. $1, -1, 1, -1, \dots, (-1)^{n+1}, \dots$

4. $-1, 1, -1, 1, \dots, (-1)^n, \dots$

5. $0, 2, 4, 6, 8, \dots, 2n, \dots$

6. $1, 3, 5, 7, 9, \dots, 2n+1, \dots$

7. (n) degree Taylor polynomial of $f(x)$ at $x=a$.

$$T_n(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n)}(a)(x-a)^n}{n!}$$

Final ñ

Section (6.2):-

volumes

① Disk: 

$$V_D = \pi \int_a^b r^2 dx \text{ (or } dy)$$

② washer: 

$$V_W = \pi \int_a^b (r_2^2 - r_1^2) dx \text{ (or } dy)$$

rotation axis.

Ⅰ rotation axis: $x = a$

$$x = f(y), \quad x = g(y)$$



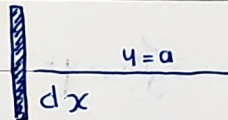
$$V = \int_c^d (r_2^2 - r_1^2) dy$$

$$x = x$$

Ⅱ rotation axis: $y = a$

$$V = \int_a^b (r_2^2 - r_1^2) dx$$

$$y = y$$



$$V_W = \pi \int_a^b (r_2^2 - r_1^2) dx \text{ (or } dy)$$

r_2 = الدالة البعيدة عن محور الدوران - معادلة محور الدوران

r_1 = الدالة القريبة من محور الدوران - معادلة محور الدوران

* إذا تم وضع أنفك ليكون الناتج موجباً

* اختياري للشريحة يكون على حسب محور الدوران ، كأن تكون الشريحة تكون عامودية على محور الدوران .

Section (8.1) + (8.2) :-

Arc length + area of surface

Def:-

- If f' cont. on $[a, b]$, then the length of the curve $y = f(x)$,

$$a \leq x \leq b \text{ is: } L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

- If $x = g(y)$, $c \leq y \leq d$ such that $g'(y)$ is cont. on $[c, d]$,

$$\text{then: } L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Summary (8.1 + 8.2) :-

• If $y = f(x)$, $a \leq x \leq b$:

- The length of the curve: $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

- Area of the curve if the curve is rotated:

$$\textcircled{a} \text{ About } x\text{-axis: } S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\textcircled{b} \text{ About } y\text{-axis: } S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

• If $x = g(y)$, $c \leq y \leq d$:

- The length of the curve: $L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

- Area of the curve if the curve is rotated:

$$\textcircled{a} \text{ About } x\text{-axis: } S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$\textcircled{b} \text{ About } y\text{-axis: } S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Section (8.3) :-

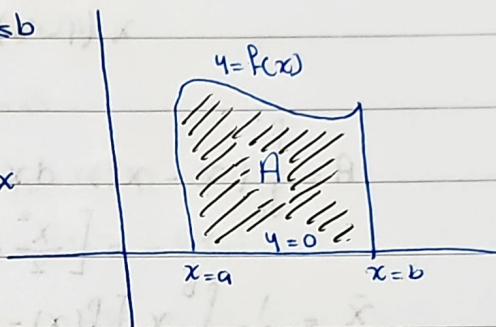
Moments and centers of mass :-

→ Centroid
= $(\bar{x}, \bar{y})$

□ $y = f(x)$, $y = 0$ (x-axis), $a \leq x \leq b$

$$A = \int_a^b f(x) dx$$

$$\bar{x} = \frac{1}{A} \int_a^b x f(x) dx, \quad \bar{y} = \frac{1}{2A} \int_a^b (f(x))^2 dx$$

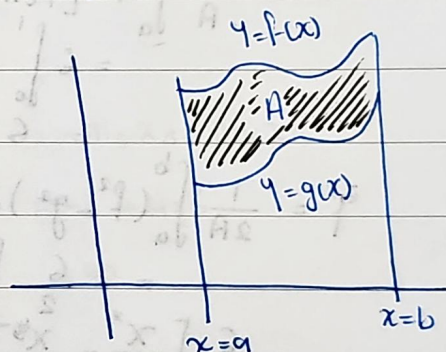


□ $y = f(x)$, $y = g(x)$, $a \leq x \leq b$

$$A = \int_a^b [f(x) - g(x)] dx$$

$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx$$

$$\bar{y} = \frac{1}{2A} \int_a^b [f^2 - g^2] dx$$



Def:

• If a region R lies between two curves $y = f(x)$, $y = g(x)$ where

$f(x) \geq g(x)$. The centroid $(\bar{x}, \bar{y})$ is given by:

$$\bar{x} = \frac{1}{A} \int_a^b x (f(x) - g(x)) dx, \quad \bar{y} = \frac{1}{2A} \int_a^b [(f(x))^2 - (g(x))^2] dx$$

$$\text{where } A = \int_a^b [f(x) - g(x)] dx$$

• If a region R lies between two curves $x = f(y)$, $x = g(y)$ where

$f(y) \geq g(y)$. The centroid $(\bar{x}, \bar{y})$ is given by:

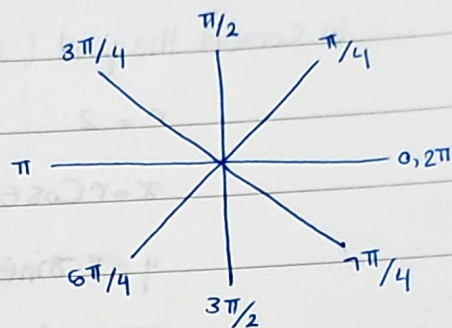
$$\bar{x} = \frac{1}{2A} \int_c^d [(f(y))^2 - (g(y))^2] dy, \quad \bar{y} = \frac{1}{A} \int_c^d y (f(y) - g(y)) dy$$

$$\text{where } A = \int_c^d [f(y) - g(y)] dy$$

Section (10.3):-

Polar Coordinates:- (r, θ)

Def: A polar curve $r = f(\theta)$ is symmetric about the polar axis if $f(-\theta) = f(\theta)$
($r(-\theta) = r(\theta)$).



حالات ثمانية :-

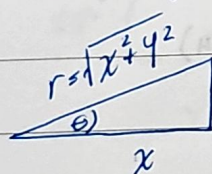
① عندما تكون r تكون r ثابتة يجب أن تكون r ثابتة مع r ثابتة r ثابتة

فقط وإضافته π على الزاوية θ

② عندما تكون الزاوية $\theta = 90^\circ$ $(0, \pi/2, \pi, 3\pi/2, 2\pi)$

③ عندما تكون الزاوية $\theta = 45^\circ$ $(0, \pi/4, \pi/2, 3\pi/4, \pi, 5\pi/4, 3\pi/2, 7\pi/4, 2\pi)$

④ Rose : $\cos(n\theta) = 2n \text{ petals}$, $\sin(n\theta) = n \text{ petals}$



$$\sin(\theta) = \frac{y}{r} \rightarrow y = r \sin(\theta)$$

$$\cos(\theta) = \frac{x}{r} \rightarrow x = r \cos(\theta)$$

- To convert from polar to Cartesian: $(r, \theta) \rightarrow (x, y)$:

$$x = r \cos \theta \quad , \quad y = r \sin(\theta)$$

- To convert from Cartesian to polar: $(x, y) \rightarrow (r, \theta)$:

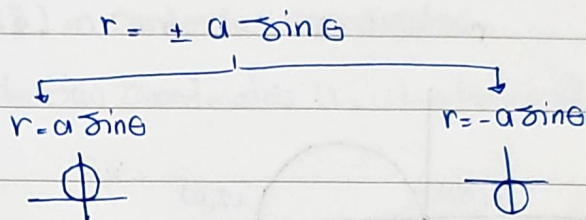
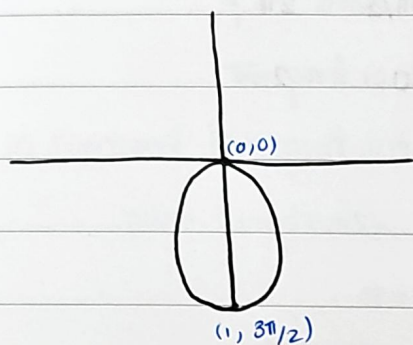
$$r^2 = x^2 + y^2 \quad , \quad \tan \theta = \frac{y}{x}$$

θ	A
$\pi - \theta$	C

$\pi - \theta$	θ
$\pi + \theta$	$\theta - 2\pi$

Ex $r = -\sin\theta$

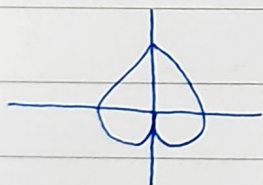
θ	0	$\pi/2$	π	$3\pi/2$	2π
r	0	-1	0	1	0



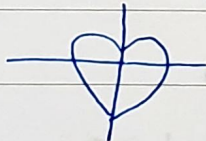
cardioids (heart shape):

$r = a \pm a \sin\theta$

$r = a + a \sin\theta$
 $r = a(1 + \sin\theta)$

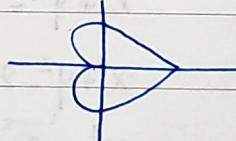


$r = a - a \sin\theta$
 $r = a(1 - \sin\theta)$

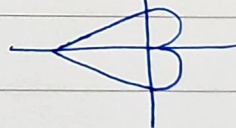


$r = a \pm a \cos\theta$

$r = a + a \cos\theta$
 $r = a(1 + \cos\theta)$

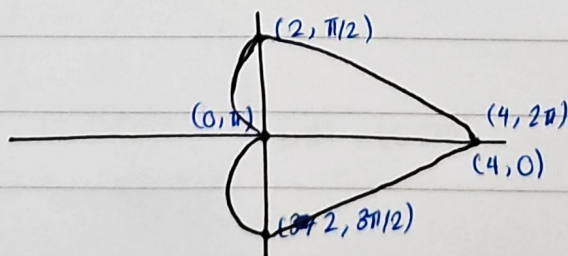


$r = a - a \cos\theta$
 $r = a(1 - \cos\theta)$



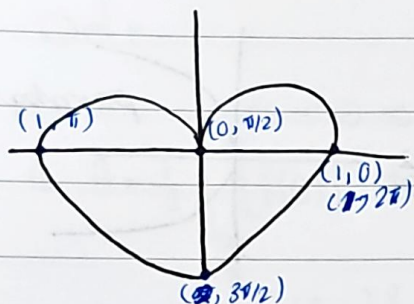
Ex Sketch: $r = 2 + 2\cos\theta$

θ	0	$\pi/2$	π	$3\pi/2$	2π
r	4	2	0	2	4

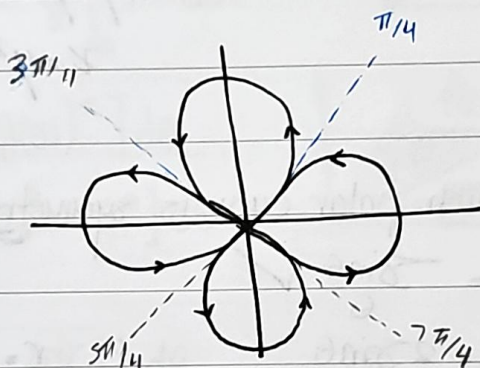


17] sketch: $r = 1 - \sin \theta$

θ	0	$\pi/2$	π	$3\pi/2$	2π
r	1	0	1	2	1



18] sketch: $r = \cos 2\theta$



19] * Tangent or slopes of polar curves:

→ If $r = f(\theta)$, (θ) is parameter and its parametric eq's are:

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$y' = \frac{dy/d\theta}{dx/d\theta} = \frac{dr/d\theta \sin \theta + r \cos \theta}{dr/d\theta \cos \theta - r \sin \theta}$$

20] Identify the curve by finding cartesian coordinates:

$$x^2 - 2x + y^2 - 2y = 0$$

$$\underbrace{x^2 - 2x + 1 - 1} + \underbrace{y^2 - 2y + 1 - 1} = 0$$

$$\underbrace{(x-1)^2 + (y-1)^2}_{\text{circle}} = 2$$

Circle with center: $(1, 1)$, $r = \sqrt{2}$

$$b) r = \csc \theta \cot \theta$$

$$r = \frac{1}{\sin \theta} \cdot \frac{x}{y}$$

$$r = \frac{x}{y \sin \theta}$$

$$x = r y \sin \theta$$

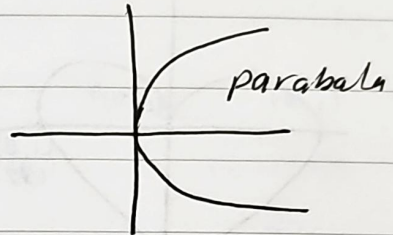
$$x = y r \sin \theta$$

$$x = y \cdot y = y^2$$

$$x = y^2$$

$$\therefore \tan \theta = \frac{y}{x}$$

$$\therefore \cot \theta = \frac{x}{y}$$



110] Which polar curve is symmetric about the polar axis?

a) $r = \frac{\sin \theta}{\theta}$ ✓

$$r(-\theta) = r(\theta)$$

b) $r = 2 \sin \theta$

$$r(-\theta) = \frac{\sin(-\theta)}{-\theta}$$

c) $r = 2 \sin \theta + 3 \cos \theta$

$$= \frac{-\sin \theta}{-\theta} = \frac{\sin \theta}{\theta} = r(\theta)$$

d) $r = 1 + \sin \theta$

Section (10.4):-

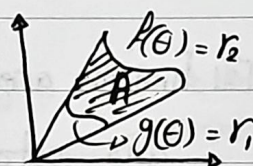
Areas and lengths in polar coordinates:-

Def:-

$$\text{I} \quad A = \frac{1}{2} \int_a^b [f(\theta)]^2 d\theta = \frac{1}{2} \int_a^b r^2 d\theta \quad \text{where } a \leq \theta \leq b, \quad r = f(\theta)$$



$$\text{II} \quad A = \frac{1}{2} \int_a^b [(f(\theta))^2 - (g(\theta))^2] d\theta = \frac{1}{2} \int_a^b [r_2^2 - r_1^2] d\theta$$



$$\text{Arc length: } L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

III Find the area enclosed by one loop of the 4-leaved rose

$$r = \cos(2\theta)$$

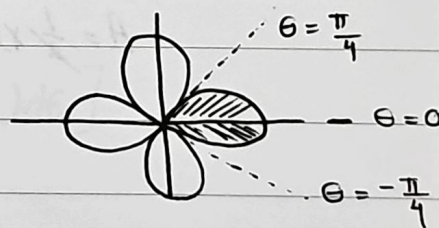
$$r=0, \quad \cos(2\theta)=0, \quad 2\theta = \cos^{-1}(0)$$

$$2\theta = \frac{\pi}{2}, \quad \theta = \frac{\pi}{4}$$

$$A = \frac{1}{2} \int_a^b r^2 d\theta = \frac{1}{2} \int_{-\pi/4}^{\pi/4} [\cos(2\theta)]^2 d\theta$$

$$= \frac{1}{2} \times 2 \int_0^{\pi/4} \left[\frac{1}{2} (1 + \cos(4\theta)) \right]^2 d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{\sin(4\theta)}{4} \right]_0^{\pi/4} = \frac{1}{2} \frac{\pi}{4} = \frac{\pi}{8}$$



حلا خطية معينة :

١) معرفة نقاط التقاطع - نضع $(r=r)$.

في حالة متبني واحد $(r=0)$.

٢) المعرفة البعيد والقريب : (r_2, r_1) - تأخذ سهم من نقطتي الأقطاب باتجاه المساحة المصدرة .

اسم الأقرب هو (r_2) ، اسم الأبعد هو (r_1) .

2] Find the area of the region that lies : Inside $r=3\cos\theta$ and outside $r=1+\cos\theta$

$$r=r$$

$$1+\cos\theta = 3\cos\theta$$

$$1 = 3\cos\theta - \cos\theta = 2\cos\theta$$

$$\cos\theta = 1/2 \rightarrow \theta = \pi/3$$

$$\begin{aligned} A &= \frac{1}{2} \times 2 \int_0^{\pi/3} (r_2^2 - r_1^2) d\theta \\ &= \int_0^{\pi/3} [(3\cos\theta)^2 - (1+\cos\theta)^2] d\theta \end{aligned}$$

